Predicting the Energy Used by a House

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Abstract

The data set used in this project is from the study conducted by Candanedo et al. [1].The data set consists of the appliance energy consumption in a single house, along with various environmental and weather related variables. The goal of this project is to develop a good predictive model for energy consumption, as well as to understand the factors that affect the energy consumption in the house. The authors of the original paper used the following methods for the prediction model: Multiple Linear Regression, Support Vector Machines, Random Forest and Gradient Boosting Machines. Among these methods, Gradient Boosting obtained the best out-of-sample performance, with an R^2 of 0.57. On the other hand Multiple Linear Regression obtained the worst out-of-sample performance with an R^2 of 0.16.

The model developed from applying ordinary regression directly to the raw 10-minute data failed on several of the key assumptions: the errors were strongly correlated, their variance was non-constant, and their distribution deviated considerably from normality. As a result, the model performed poorly, with an out-of-sample R^2 of only 0.09. However, our main objective was not to simply obtain a model with good predictive accuracy, but also one that was interpretable.

We worked our way with two alternative approaches. The first was a classical time-series approach using an autoregressive (AR) model. The model predicted energy consumption over the next 10 minutes reasonably well but it relied exclusively on the past values of the response variable, and its predictive advantage over simply using the mean disappeared after a few hours. In the second approach, we aggregated the observations into daily averages, allowing each day to be treated approximately as an independent observation. We then applied ordinary least squares regression combined with variable selection. Using backward selection with the Bayesian Information Criterion (BIC), we obtained a model with seven predictors: the daily averages of four room temperatures, two humidity measurements and lighting consumption. This model achieved an out-of-sample R^2 of 0.34 under random splits and 0.20 when evaluated on later days using a time-ordered split.

Furthermore, we tested whether comparable predictions could be obtained when the house reports only one set of measurements per day. We found that a reading taken at 6 PM provided the best performance among the times considered, achieving an R^2 of 0.58, while a 3 PM reading gave the weakest performance. These results highlight the potential of daily aggregation and carefully selected predictors for developing interpretable models of household energy consumption.

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1 Dataset

1.1 Where the data comes from

The data is the "Appliances energy prediction" data set from the UCI repository, collected by Candanedo, Feldheim and Deramaix [1]. It comes from one low-energy house. As described in the paper, temperature and humidity are measured by a wireless sensor network placed in different rooms, the energy used by appliances and by lights comes from energy meters, and the outside weather comes from the nearest airport weather station and is joined to the house data by the time stamp. Everything is given as one row every 10 minutes, from 11 January 2016 17:00 to 27 May 2016 18:00, so about 4.5 months.

The file has 19 735 rows and 29 columns. All data were used for the analysis, there are no missing values, there are no duplicated rows, there are no constant columns, and all gaps between rows are of 10 minutes. The first and the last day of the data set are not complete and consist of 42 and 109 readings, respectively. Therefore, we used 136 full days of 144 readings.

Variable	Meaning
`Appliances`	**Response**: energy used by appliances, Wh per 10 minutes
`lights`	Energy used by the lights, Wh
`T1`–`T9`	Temperature (°C) in kitchen, living room, laundry room, office, bathroom, outside (north side), ironing room, teenager's room, parents' room
`RH_1`–`RH_9`	Humidity (%) at the same nine places
`T_out`, `RH_out`, `Press_mm_hg`, `Windspeed`, `Visibility`, `Tdewpoint`	Weather station: temperature, humidity, pressure, wind speed, visibility, dew point
`rv1`, `rv2`	Two random numbers added by the authors (we drop them)

Table 1: Variables in the data. We use the 25 covariates (everything except `Appliances`, `rv1`, `rv2`).

1.2 The response and why we do not remove outliers

`Appliances` has mean 97.7, median 60 and standard deviation 102.5. It goes from 10 to 1080 Wh and it only takes values in steps of 10 (92 different values). It is strongly right skewed (skewness 3.39, and 1.17 after taking logs). `lights` is zero in 77% of the readings.

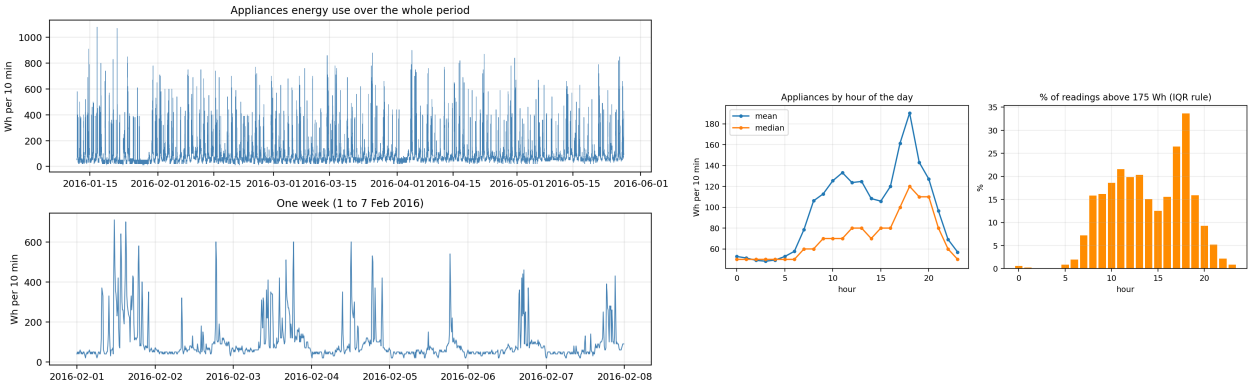


Figure 1: Left: Appliances over the whole period and over one week. Right: mean and median by hour of the day, and the share of readings above the IQR fence (175 Wh).

Figure 1 shows a low base level of about 50 Wh with tall short spikes, and a clear daily pattern:

the mean is about 50 at night, rises in the morning (133 at 11 am) and has its maximum of 190 at 6 pm. By the IQR rule ($Q_1 = 50$, $Q_3 = 100$, fence = 175) 10.8% of the readings are outliers. But they are not random. They are mostly between 8 am and 1 pm and between 5 and 7 pm (33.7% of the readings at 6 pm are above the fence, against less than 1% at night). They look like real use of the house (we think cooking, washing and so on), so they are the thing we want to predict and we do not delete them. When we average over days later, the spikes also get smoothed out, so we do not go further with outlier removal.

1.3 It is a time series

Readings next to each other are not independent. In Figure 2 the ACF of `Appliances` is 0.75 at lag 1, 0.53 at lag 2, goes down slowly (0.07 at lag 30) and is still 0.22 at lag 144, which is one day. The PACF plot of the appliances time-series has one big spike at lag 1 and then only small values after that. So the series behaves like an autoregressive process, and the independence assumption of ordinary regression is in doubt from the start.

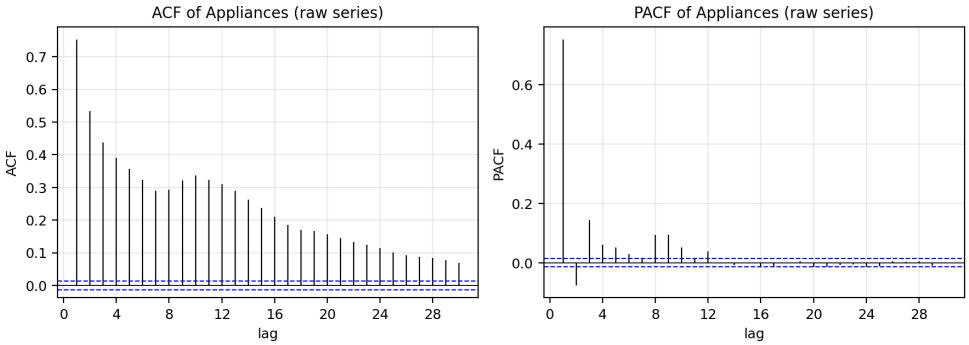


Figure 2: ACF and PACF of `Appliances`. The dashed lines are $\pm 1.96/\sqrt{n}$.

1.4 Why we drop `rv1` and `rv2`

The two columns are not exactly the same but they are random numbers added by the authors to test models. Their correlation with `Appliances` is -0.011 (p-value 0.117), so there is no linear relationship, which fits with them being noise. We drop them. Other covariates also have small correlations with the response, but we keep those, because a variable can matter once the others are in the model. Which ones are really needed is decided later by model selection.

1.5 Correlations and collinearity

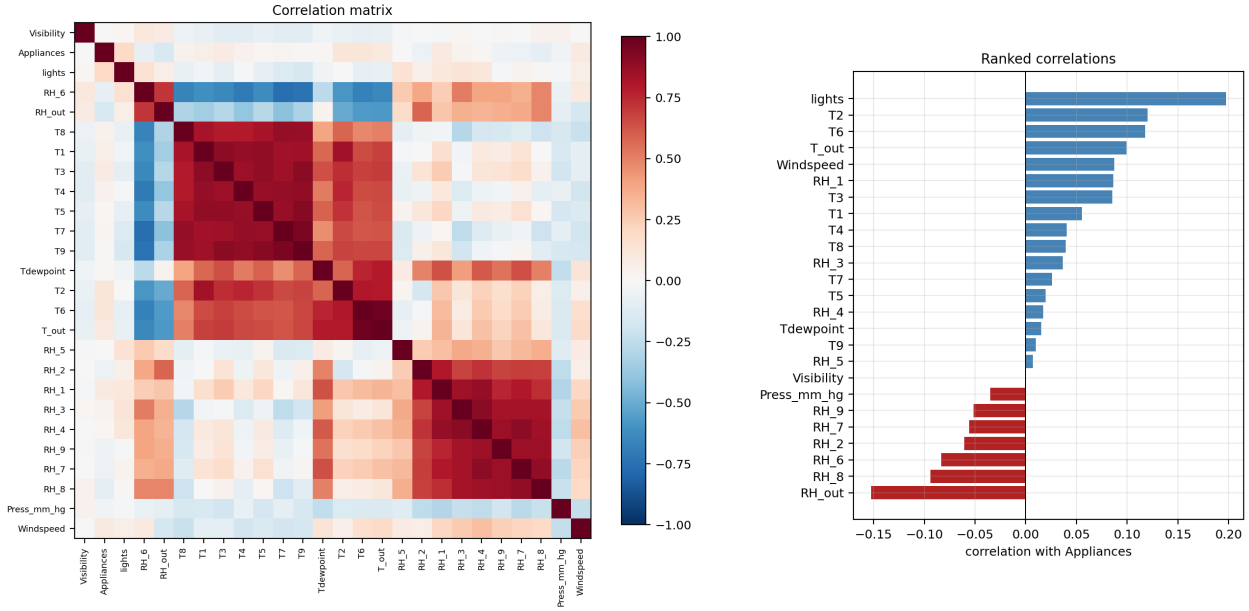


Figure 3: Left: correlation matrix (variables ordered by similarity). Right: correlation of each covariate with `Appliances`.

The row of `Appliances` in the correlation matrix is almost white. The largest correlations with the response are `lights` (0.197), `RH_out` (−0.152), `T2` (0.120) and `T6` (0.118), and none is above 0.2, so each covariate alone explains very little. On the other hand the covariates are strongly related with each other: the indoor temperatures form one block, the indoor humidities another, and `RH_6` and `RH_out` are negatively related to the temperatures. Some pairs are almost the same variable: `T6`–`T_out` (0.975), `T7`–`T9` (0.945), `T5`–`T9` (0.911), `T3`–`T9` (0.901).

The variance inflation factor of covariate j is $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing x_j on the other covariates. For the 10 minute data (Figure 4) the largest VIF is 234 (`T_out`) and 13 of the 25 are above 10, and the condition number of the standardized design matrix is 62. This is serious collinearity. It does not bias OLS, but it makes the variance of the estimates large and the coefficients unstable, and we will see it in the base model.

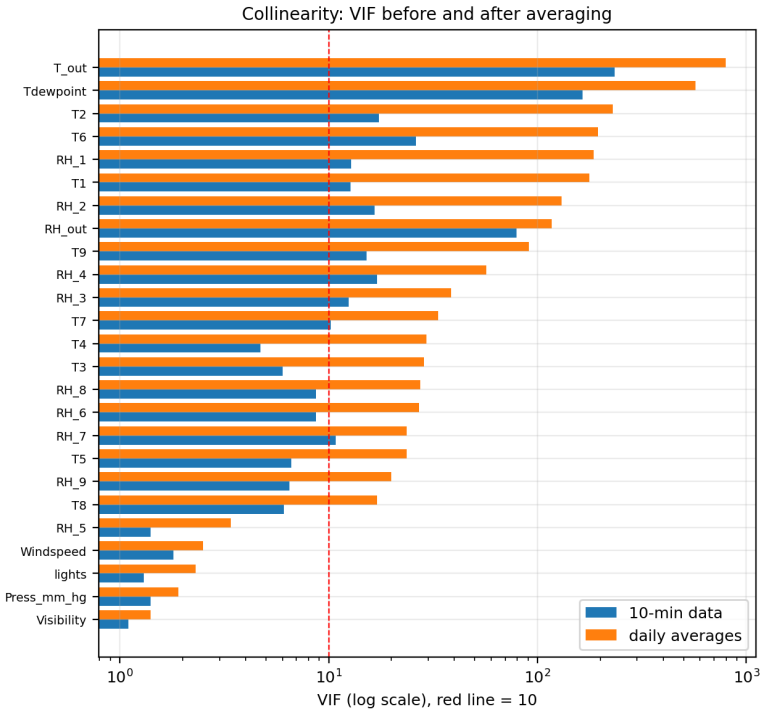


Figure 4: VIF of every covariate for the 10 minute data and for the daily averages (log scale, red line at 10). The daily averages are used in Section 3.2.

2 Base model

2.1 Model and fit

The first thing to try is a plain multiple linear regression on the 10 minute data:

$$\text{Appliances}_t = \beta_0 + \sum_{j=1}^{25} \beta_j \, x_{tj} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma^2) \text{ independent.}$$

Since the data is time series we do not shuffle it. The first 80% (15 788 readings, until 30 April) is the training part and the last 20% (3 947 readings, until 27 May) is the test part. Table 2 gives the fitted model. It has $R^2 = 0.173$, adjusted $R^2 = 0.172$, $F = 131.8$ and $\hat{\sigma} = 95.75$.

	Estimate	t	marg. r		Estimate	t	marg. r
Intercept	−67.29	−0.59	–	`RH_5`	0.106	1.04	0.007
`T1`†	−6.37	−2.85	0.055	`RH_6`†	0.342	4.22	−0.083
`T2`†	−13.21	−6.30	0.120	`RH_7`	−0.489	−0.99	−0.056
`T3`	27.71	22.28	0.085	`RH_8`	−5.96	−13.88	−0.094
`T4`†	−3.80	−3.47	0.040	`RH_9`	−1.00	−2.10	−0.052
`T5`	−1.68	−1.11	0.020	`T_out`†	−8.63	−3.05	0.099
`T6`	7.64	9.15	0.118	`Press_mm_hg`	0.223	1.90	−0.035
`T7`	1.08	0.70	0.026	`RH_out`	−0.521	−0.99	−0.152
`T8`	9.04	8.17	0.040	`Windspeed`	1.27	3.21	0.087
`T9`†	−14.35	−6.89	0.010	`Visibility`	0.190	3.01	0.000
`RH_1`	12.85	16.95	0.086	`Tdewpoint`	3.70	1.33	0.015
`RH_2`	−12.41	−13.60	−0.061	`lights`	2.14	20.43	0.197
`RH_3`	8.63	10.27	0.036				
`RH_4`†	−2.45	−3.33	0.017				

Table 2: Base model, fitted on the training part. "marg. r" is the correlation of the covariate with `Appliances`. † significant coefficient with the opposite sign of the marginal correlation.

Seven significant coefficients have the opposite sign of their marginal correlation (for example `T2` has $r = +0.12$ but coefficient -13.2, and `T6` and `T_out`, which have correlation 0.975, get $+7.6$ and -8.6).

2.2 Accuracy on the test part

	RMSE	MAE	R^2_{oos}
Base OLS (25 covariates)	86.73	52.01	0.093
Benchmark (always predict the training mean)	91.05	52.69	0

Table 3: Out of sample results of the base model (last 20% of the data).

Here $R^2_{\text{oos}} = 1 - \sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y}_{\text{train}})^2$, so it compares the model with the benchmark that always predicts the training mean. The base model is only 4.7% better than the benchmark in RMSE, the MAE is almost the same, and the R^2_{oos} of 0.093 is about half of the in-sample 0.173. So it predicts badly.

2.3 Violated assumptions

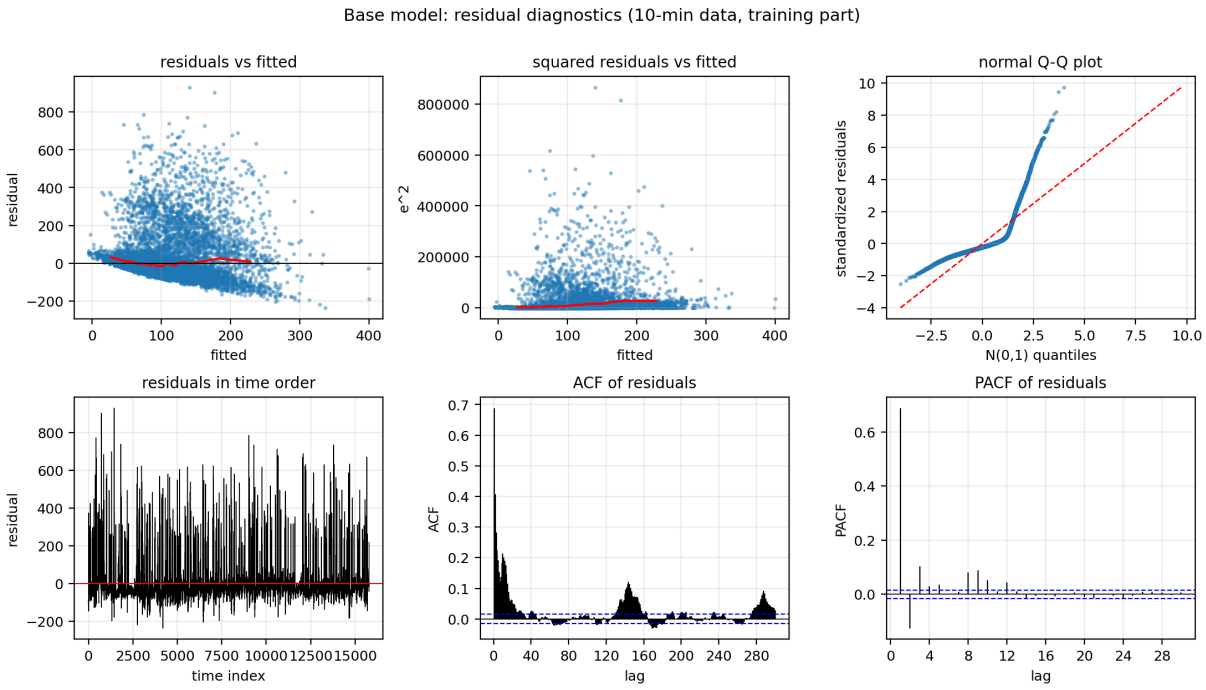


Figure 5: Residual plots of the base model (training part).

Figure 5 shows the residuals of the base model. After going through the assumptions one by one, we look at the results of the tests: D is the Durbin–Watson statistic, S the Breusch–Pagan statistic (regression of $e_i^2/\hat{\sigma}^2_{ML}$ on the covariates) and S' the Cook–Weisberg statistic (the same regression on the fitted values only).

Constant variance (heteroscedasticity). In the plot of residuals against fitted values the points make a pattern. The spread grows with the fitted value, and the plot of e_i^2 against the fitted values goes up as well. Breusch–Pagan gives $S = 4648$ on 25 df and Cook–Weisberg $S' = 3980$ on 1 df, both with p-value ≈ 0. So the variance is not constant, and it depends on the level of the fitted energy.

Normality. The Q–Q plot bends up sharply on the right side: standardized residuals go up to about 10, where a normal sample would reach about 3.5. The residual skewness is 3.28 and the kurtosis 17.2 (normal: 0 and 3). The reason is the spikes, which the model cannot explain. The normal assumption fails.

Autocorrelation and independence. The residuals are in time order and show long groups of large positive values. The ACF starts at 0.69 at lag 1 and goes down slowly, with new bumps near lags 144 and 288, which are exactly one and two days. The PACF shows a spike at lag 1 (0.69), a negative value at lag 2 and then small values. Durbin–Watson for the residuals give $D = 0.622$, hence the lag 1 correlation is $r_1 = 0.689$. The errors are strongly dependent, and the daily bumps tell us that the model is completely insensitive to the time of the day in it.

Consequences. Because of the dependence the standard errors are too small and therefore the t-values in Table 2 and the F-test are not reliable and we use them only as description. The collinearity from Section 1.5 gives the sign changes. This is why the model does not work, and it gives us two ways to go on.

3 Our solutions to the problems

3.1 Method 1: time series (AR) model

If the main problem is the dependence in time, the natural model is a time series model. We use an autoregressive model of order p:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t,$$

fitted on the same training part, and tested on the same last 20%. The order p is chosen with AIC (we also look at the PACF). The AIC curve in Figure 6 falls a lot until order 10 and then it is flat; the minimum is at $p = 14$ (BIC would give 12), so we take AR(14). The first coefficients are $\hat{c} = 13.81$, $\hat{\phi}_1 = 0.794$, $\hat{\phi}_2 = -0.179$, $\hat{\phi}_3 = 0.084$, $\hat{\phi}_4 = 0.013$, $\hat{\phi}_5 = 0.026$. This agrees with the PACF, which has a big spike at lag 1 and small values afterwards.

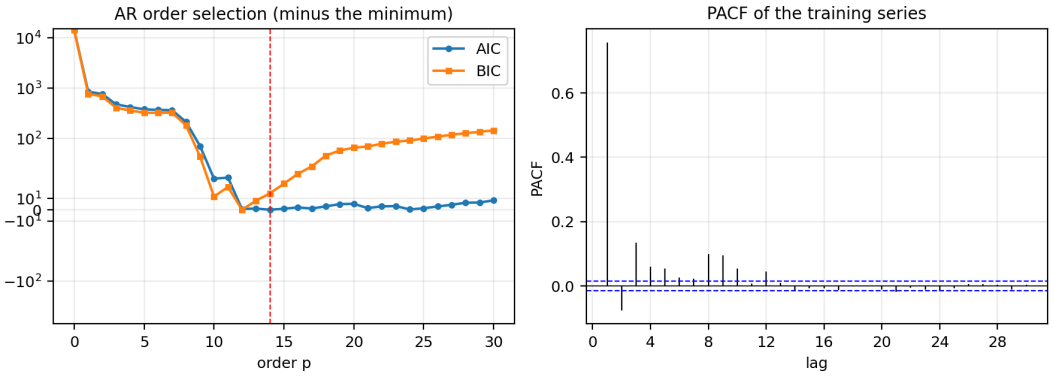


Figure 6: Left: AIC and BIC against the AR order (minus their minimum, symlog scale). Right: PACF of the training series.

One step ahead (10 minutes)	RMSE	MAE	R^2_{oos}
AR(14)	60.03	26.97	0.565
Naive (previous reading)	66.44	26.65	0.468
Benchmark (training mean)	91.05	52.69	0

Table 4: AR(14) forecasts on the last 20% of the data, using the true previous readings.

For the next 10 minutes the AR model is much better than the regression (RMSE 60.0 against 86.7) and 9.6% better than the naive forecast. The residuals of the AR model have $D = 2.05$ and $r_1 = -0.026$ (p-value 0.099 with the large sample test), so the lag 1 dependence is gone. Still, the ACF of its errors has a bump of 0.087 at lag 144, so the daily cycle is not captured.

Problems with this model.

- It only uses the past of `Appliances`. No temperature, humidity or lights appears in it, so it says nothing about what the energy depends on.
- To forecast the next 10 minutes it needs the last 14 readings, which is 140 minutes of data from the same house. If we do not have that, we cannot use it.
- It loses its power quickly when we look further ahead (Figure 7). The RMSE is 60 for 10 minutes ahead, 78 for 30 minutes, 82 for one hour and 87.5 for six hours, where it is the same as predicting the mean. The naive forecast is even worse than the mean after about 20 minutes.
- The MAE of the naive forecast is a slightly smaller than that of AR(14) (26.65 against 26.97). Thus, for the typical reading the gain over repeating the last value is small. The gain is in the large values.

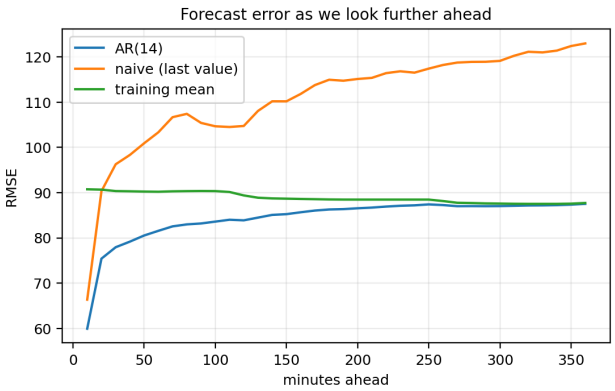


Figure 7: RMSE of the AR(14) forecast against how far ahead we forecast (forecasts built from earlier forecasts).

So the time series model is good for predicting the immediate next reading but not for what the energy depends upon which was one of our goals. We stop here with it, since a better time series model would need more than we cover in the course and not exactly what we are looking for.

3.2 Method 2: daily averages

3.2.1 Idea and model

Instead of predicting the energy usage of a single reading 10 minutes into the future, we can try to predict the average energy usage of a day based on the average values of the same variables of that day. Averaging 144 readings removes most of the dependence between neighbouring readings, so each day can be treated as nearly one independent observation. We keep only the 136 full days (they are consecutive days, from 12 January to 26 May) and for each day d we take

$$\bar{y}_d = \frac{1}{144}\sum_{t=1}^{144} \text{Appliances}_{d,t}, \qquad \bar{x}_{dj} = \frac{1}{144}\sum_{t=1}^{144} x_{d,t,j}.$$

The daily model is $\bar{y}_d = \beta_0 + \sum_{j=1}^{25} \beta_j \bar{x}_{dj} + \varepsilon_d$, $d = 1, \ldots, 136$. The daily response has mean 97.4, standard deviation 30.3 and is much less skewed (0.84, and 0.03 on the log scale). Since the daily

mean is just the daily total divided by 144, modelling the mean or the total gives the same R^2 and t-values. Covariates are averaged and never summed.

3.2.2 Before and after averaging

	10 minute base model	Daily model (25 covariates)
n	15 788	136
Adjusted R^2	0.172	0.444
Lag 1 residual autocorrelation r_1	0.689	0.076
Durbin–Watson D	0.622	1.843
Durbin–Watson test, p-value (large sample)	≈ 0	0.373
Breusch–Pagan p-value	≈ 0	0.0085
Cook–Weisberg p-value	≈ 0	0.0007
Residual skewness / kurtosis	3.28 / 17.19	0.43 / 3.41
Skewness of response (raw / log)	3.32 / 1.13	0.84 / 0.03
Largest VIF	234	798
Number of VIF above 10	13	20
Condition number	62.2	126.4

Table 5: The same checks for the 10 minute base model and for the daily model.

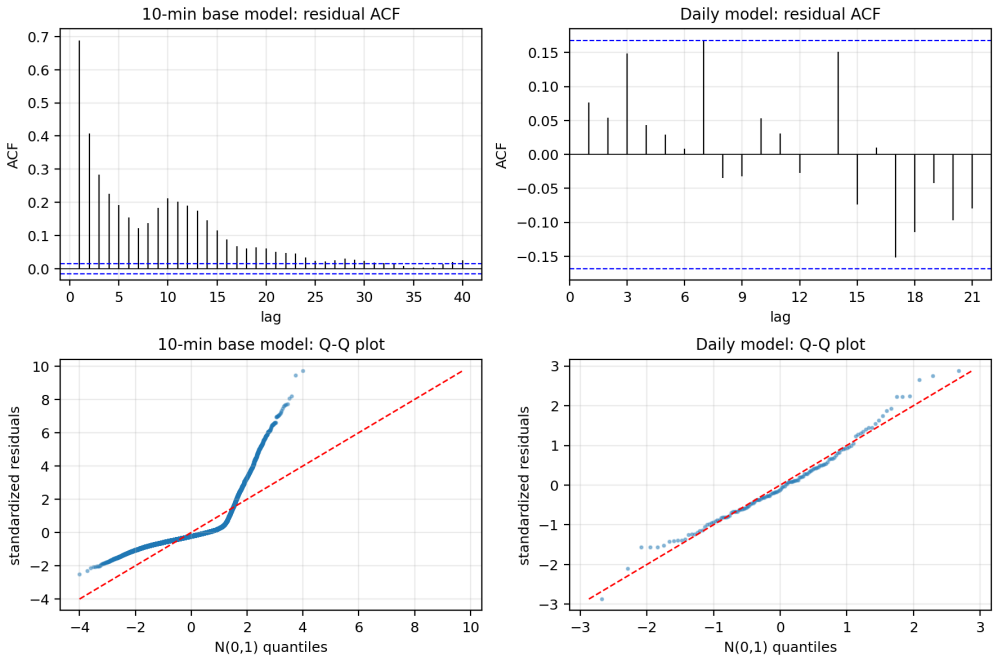


Figure 8: Residual ACF and Q–Q plot, before (10 minute data) and after (daily data) averaging.

What changed (Table 5 and Figure 8):

- **Autocorrelation** is much smaller. The lag 1 correlation goes from 0.689 to 0.076, D from 0.62 to 1.84, and all the bars of the daily residual ACF are inside the band $\pm 1.96/\sqrt{136} = \pm 0.168$ (lag 7 is exactly on its edge, a small hint of a weekly pattern). The large sample Durbin–Watson test does not reject any more ($z = 0.89$, p-value 0.37). We still would not say it is completely gone: an exact Durbin–Watson test one-sided against positive correlation, which we ran before in R, gave p-value 0.017 for the same D. So the dependence is small but maybe not zero.

- **Normality** has improved drastically. The Q–Q plot is almost on the line, and skewness and kurtosis are close to 0.43 and 3.4. The huge right tail of the 10 minute residuals is gone.
- **Constant variance** is better but not solved. The test statistics are much smaller (Breusch–Pagan $S = 44.9$ against 4648, both on 25 df) but the p-values (0.0085 and 0.0007) still reject with $n = 136$. We come back to this in Section 3.2.5.
- **Collinearity** gets worse (VIF up to 798, condition number 126, Figure 4). This might be because the daily average, while removing the independent fluctuations, keeps the slow changes of the weather that affect all temperatures and humidities simultaneously. .

3.2.3 Model building and selection

There are $2^{25} \approx 33.5$ million possible subsets of the 25 covariates, so all possible regressions is not practical. We use the idea of forward and backward stepwise regression, which give one nested sequence of models (sizes $k = 0, \ldots, 25$), and we compare the sizes with

$$\bar{R}^2_k = 1 - \frac{\mathrm{RSS}_k/(n-k-1)}{\mathrm{SST}/(n-1)}, \quad C_p = \frac{\mathrm{RSS}_k}{\hat{\sigma}^2_F} + 2(k+1) - n, \quad \mathrm{AIC} = n \log \frac{\mathrm{RSS}_k}{n} + 2(k+1), \quad \mathrm{BIC} = n \log \frac{\mathrm{RSS}_k}{n} + (k+1) \log n,$$

and with leave-one-out cross validation, $\mathrm{LOOCV} = \frac{1}{n} \sum_i \big(e_i/(1-h_{ii})\big)^2$. A larger $\bar{R}^2$ and smaller C_p (close to $k+1$), AIC, BIC and LOOCV are better. Here $\hat{\sigma}^2_F$ is the variance estimate from the full model.

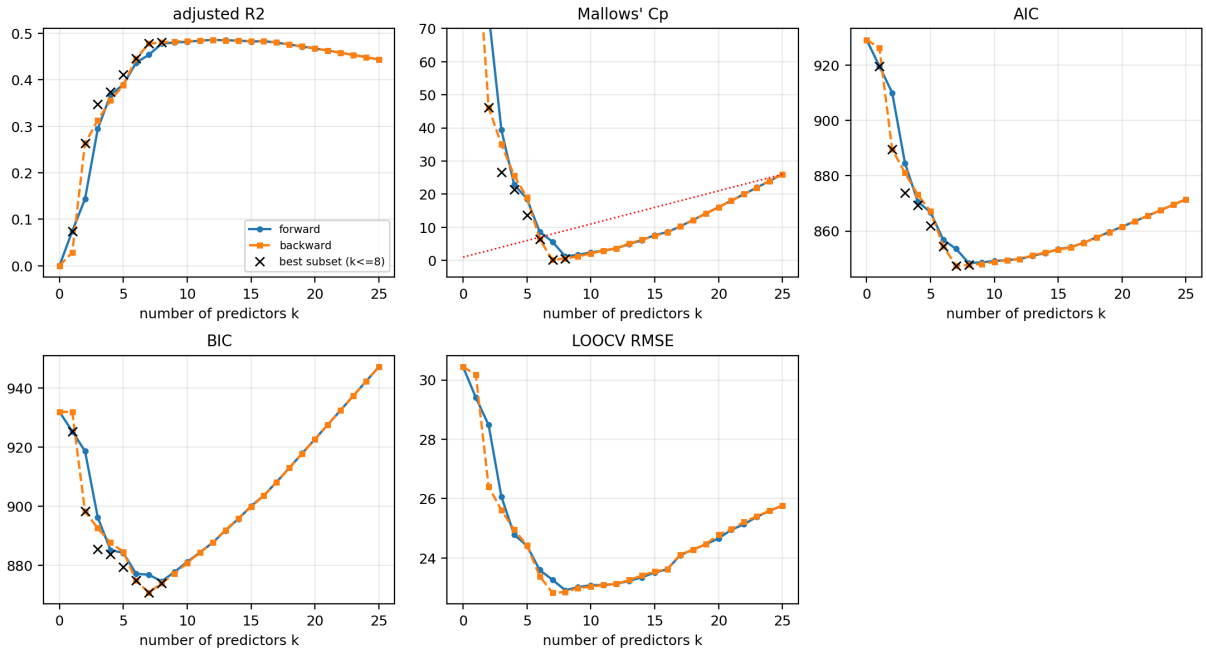


Figure 9: Criteria against the number of predictors k for forward and backward selection. The crosses are the best subset of each size (all possible regressions) for $k \leq 8$. The dotted line in the C_p plot is $C_p = k + 1$.

Method	Criterion	Chosen model
Forward	$\bar{R}^2$	12 variables: `T3, T4, T5, T7, T8, T9, RH_1, RH_2, RH_6, RH_7, T_out, lights`
Forward	C_p, AIC, BIC, LOOCV	8 variables: `T3, T4, T7, T9, RH_1, RH_7, T_out, lights`
Backward	$\bar{R}^2$	the same 12 variables
Backward	C_p, AIC, BIC, LOOCV	**7 variables**: `T3, T4, T7, T9, RH_1, RH_7, lights`

Table 6: Model chosen by each criterion (136 days).

Adjusted R^2 is the weakest penalty and picks 12 variables, but it is almost flat after 7 (Figure 9). C_p, AIC, BIC and LOOCV all choose 7 variables for backward and 8 for forward, where the only difference is that forward also puts `T_out`. The BIC of the backward model is 870.78 and of the forward one 874.62, so the backward model is better by 3.85. As a check we also ran all possible regressions for every size up to 8. The best subset with 7 variables is exactly the backward model (the same residual sum of squares, 61 479), while forward selection finds the best subset only for $k = 1$. That is why we take the **backward 7 variable model**.

We also tested the choice with F-tests for nested models (computed from the residual sums of squares). Compared with the full 25 variable model, the 7 variable model does not fit significantly worse ($F = 0.56$ on 18 and 110 df, p-value 0.92), so the 18 dropped variables are not needed. And adding `T_out` to the 7 variables (the forward model) does not help ($F = 1.00$ on 1 and 127 df, p-value 0.32).

3.2.4 Is a transformation needed? (Box–Cox)

The Box–Cox transformation is $y^{(\lambda)} = (y^\lambda - 1)/\lambda$ for $\lambda \neq 0$ and $\log y$ for $\lambda = 0$. We choose λ with the profile likelihood $\ell^*(\lambda) = C - \frac{n}{2} \log \frac{\mathrm{RSS}(\lambda)}{n} + (\lambda - 1) \sum \log y_i$ for the 7 variable model.

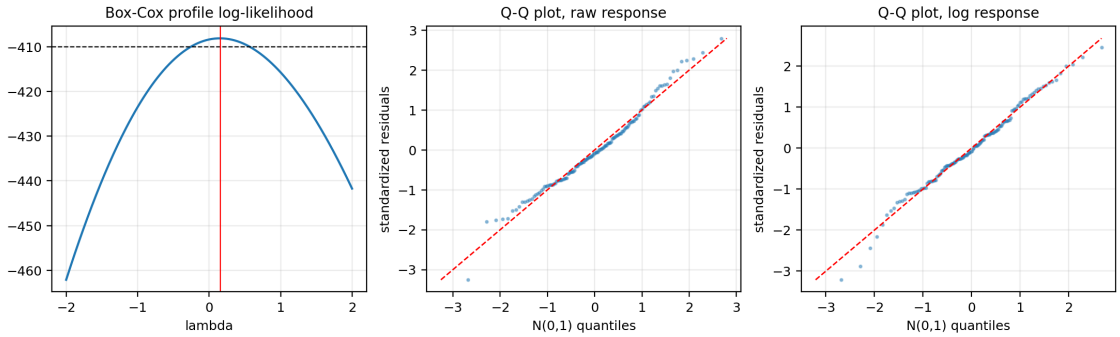


Figure 10: Box–Cox profile likelihood (the dashed line is 1.92 below the maximum) and Q–Q plots of the 7 variable model for the raw and the log response.

The maximum is at $\hat{\lambda} = 0.16$ and the approximate 95% interval is $[-0.26, 0.56]$. It contains 0 (the log) but not 1, so in principle the raw scale is not the best one. In practice the difference is small: the residual skewness is 0.32 on the raw scale and -0.15 on the log scale, and both Q–Q plots are close to the line. The log also does not fix the variance (Breusch–Pagan p-value 0.012 raw and 0.021 log). We keep the **raw scale**, because then the coefficients are directly in Wh and the evaluation is in the original units. Using $\log y$ would be a reasonable alternative.

3.2.5 Final model

The final model is

$$\widehat{\bar{y}}_d = -78.20 + 32.77\,\mathrm{T3}_d - 10.81\,\mathrm{T4}_d + 18.40\,\mathrm{T7}_d - 36.46\,\mathrm{T9}_d + 3.93\,\mathrm{RH_1}_d - 4.51\,\mathrm{RH_7}_d + 2.83\,\mathrm{lights}_d,$$

where $\bar{y}_d$ is the average `Appliances` (Wh per 10 minutes) of day d and every predictor is the daily mean of that variable. It has $R^2 = 0.505$, adjusted $R^2 = 0.478$ (the full model has 0.444), $\hat{\sigma} = 21.9$ and overall $F = 18.65$ on 7 and 128 df (p-value 5.7×10^{-17}).

	Estimate	Std. error	t	p-value	95% CI	Std. coef.
Intercept	−78.20	32.10	−2.44	0.016	[−141.7, −14.7]	97.40
`T3`	32.77	3.42	9.58	< 0.001	[26.0, 39.5]	62.43
`T4`	−10.81	3.09	−3.50	0.001	[−16.9, −4.7]	−21.02
`T7`	18.40	3.91	4.71	< 0.001	[10.7, 26.1]	37.71
`T9`	−36.46	5.45	−6.69	< 0.001	[−47.2, −25.7]	−72.98
`RH_1`	3.93	1.32	2.97	0.004	[1.3, 6.5]	14.14
`RH_7`	−4.51	1.01	−4.48	< 0.001	[−6.5, −2.5]	−20.82
`lights`	2.83	0.72	3.95	< 0.001	[1.4, 4.2]	8.93

Table 7: Final 7 variable model. "Std. coef." is the coefficient when the predictors are standardized (effect of one standard deviation).

Interpretation. Every predictor in our final model is significant, but the p-values and intervals are a bit optimistic since the variables were selected on the same data. The standardized coefficients show that `T9` (the parents' room, −73) and `T3` (the laundry room, +62) matter most, then `T7` (+38), `T4`, `RH_7` (both about −21), `RH_1` (+14) and `lights` (+9). The four temperatures have opposite signs and the sum of their standardized coefficients is only +6, which is close to zero. So the model does not say that warmer houses use more or less energy. It uses *differences* between rooms. These temperatures are strongly related (VIF inside the final model: `T9` 33, `T7` 18, `T3` 12, `T4` 10; condition number 14), so a single temperature coefficient should not be read as a physical effect. Note that T9 has a very small marginal correlation with the response variable (0.010) and yet it is the most important predictor. This is the reason why we did not use the correlation with the response variable Appliances as a criterion for variable selection.

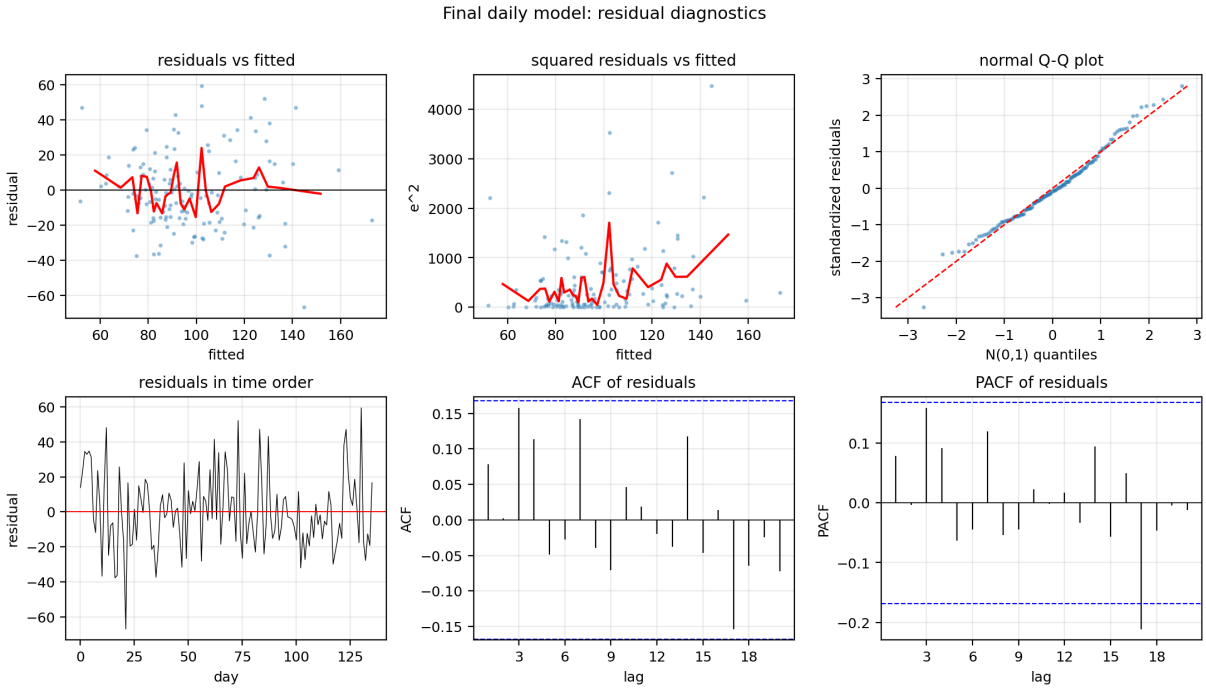


Figure 11: Residual plots of the final daily model (compare with Figure 5).

Residuals of the final model (Figure 11). Compared with the base model the picture is much better. Residuals against fitted values are a shapeless cloud and the largest standardized residual is 3.25. The Q–Q plot is close to the line (skewness 0.32, kurtosis 3.32). The residuals in time order have no long groups, and the ACF and PACF stay inside the band at almost all lags (lags 3, 7 and 14 are near the edge; the one PACF bar outside, at lag 17, is about what we expect by chance among 20 lags). $D = 1.836$ with $z = 0.91$ (p-value 0.36). Constant variance is still the weak

point: Breusch–Pagan gives $S = 17.98$ on 7 df (p-value 0.012) and Cook–Weisberg $S' = 10.22$ on 1 df (p-value 0.0014), and the plot of e^2 against the fitted values goes up a little. As a check we refitted the model by weighted least squares, with variance proportional to the square of the mean (iterated until it settled). The signs do not change and the tests no longer reject (p-values 0.41 and 0.84), but the coefficients move (for example `T3` 32.8 to 29.1, `T9` −36.5 to −30.2, `lights` 2.8 to 4.1). The heteroscedasticity is mild, so we keep OLS and take the standard errors of the temperatures and the humidities as approximate.

3.2.6 Evaluation

For the out of sample accuracy we repeated 200 random splits of the 136 days into 109 training days (80%) and 27 test days (20%). In *every* split the variable selection was done again using only the training days, so the test days never influence the model. We compare the benchmark (training mean), the full OLS, backward selection with AIC and with BIC, and forward selection with BIC.

	RMSE (sd)	MAE	R^2_{oos} (sd)	beats benchmark	average # variables
Benchmark (training mean)	30.11 (3.54)	23.53	0	–	0
Full OLS	26.08 (3.42)	20.38	0.210 (0.318)	82.5%	25
Backward AIC	24.89 (3.35)	19.35	0.285 (0.253)	88.5%	9.3
Backward BIC	**23.93** (3.40)	**18.66**	**0.339** (0.239)	**93.0%**	6.9
Forward BIC	24.95 (3.01)	19.53	0.283 (0.233)	89.5%	6.4

Table 8: Mean (and standard deviation) over 200 random 80/20 splits. "beats benchmark" is the share of splits where the RMSE is lower than the benchmark.

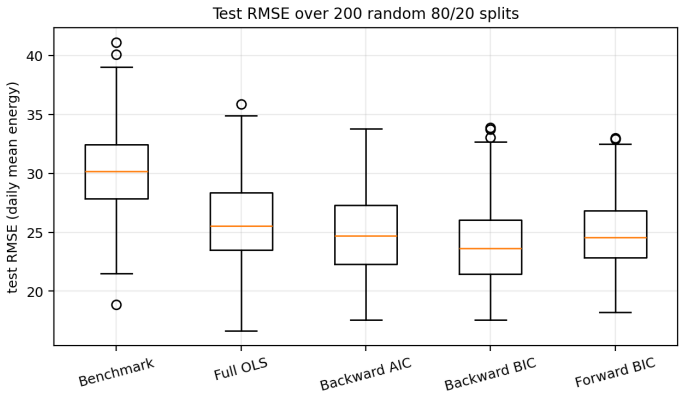


Figure 12: Test RMSE over the 200 splits.

Backward BIC is the best. Its mean RMSE is 20.5% lower than the benchmark, its R^2_{oos} is 0.34 and it beats the benchmark in 93% of the splits. The full model with 25 variables is much worse ($R^2_{\text{oos}} = 0.21$): in-sample its adjusted R^2 was 0.444, so this is the overfitting that selection removes. The model also seems stable: across the splits backward BIC selected `T9` and `T3` every time, `T7` in 95%, `lights` in 94%, `T4` in 93%, `RH_7` in 80% and `RH_1` in 69% of the splits. The differences between the selection methods are small compared with the spread over splits (standard deviation about 3.4, since a test set has only 27 days); backward BIC beats backward AIC in 65.5% and forward BIC in 68% of the splits. We did not do a formal test for these differences (the splits are not independent), so we only say that the BIC model is a bit better. The 95% prediction intervals of the BIC model contain 93.8% of the test days, which is close to the nominal level, but they are wide (about ±45 Wh, compared with a standard deviation of 30 for the daily mean).

A check in time order. The random splits treat days as independent, and we saw that this is

only roughly true. So we also trained on the first days and tested on later days: we train on days 1 to $s-1$ and test on the next 7 days, for $s = 81, 88, \ldots, 130$ (56 test days).

	RMSE	MAE	R^2_{oos}
Benchmark	30.88	24.16	0
Full OLS	31.41	23.54	−0.035
Backward AIC	28.29	20.72	0.161
Backward BIC	27.62	21.67	0.200
Forward BIC	28.42	21.81	0.152

Table 9: Time ordered evaluation on 56 test days.

The ranking is the same (full OLS is worse than doing nothing and backward BIC is the best) but all the numbers are lower: R^2_{oos} of the BIC model is 0.20 here against 0.34 with random splits. So 0.34 is probably a bit optimistic, and the real accuracy for new days is nearer to 0.2–0.3. We should also keep in mind that 56 days is a small test set.

3.2.7 Combined model: only one reading in the day

It is not realistic that a house reports every temperature at every moment. A more realistic case is that the house reports all its covariates once, at a fixed time. We tried this with 3 pm, 6 pm and 9 pm, in two steps:

1. **Estimate the daily averages.** For every covariate j we fit, on the training days, a straight line from the reading at that time to the true daily average: $\widetilde{x}_{dj} = a_j + b_j\, x_{d,j}(t_0)$.
2. **Predict the energy.** The estimated daily averages go into the daily model.

We tried two versions. In **model A** the daily model is the final model of the last section (backward BIC, fitted on the *true* daily averages), and we only feed it the estimated averages. In **model B** the same selection is done, but the model is fitted on the *estimated* averages, so it learns from the same kind of input it will get. Both are evaluated on the same 200 splits as before, and in every split the Step 1 lines and the selection use only the training days.

Reading at	Model	RMSE (sd)	R^2_{oos}	beats benchmark
	Benchmark	30.11 (3.54)	0	–
	True daily averages (all day data)	23.93 (3.40)	0.339	93%
3 pm	A	28.56 (3.86)	0.033	58%
	B	24.15 (2.74)	0.334	95%
6 pm	A	26.69 (4.16)	0.155	75%
	B	**18.99** (2.79)	**0.580**	**99%**
9 pm	A	24.02 (3.64)	0.320	92%
	B	21.78 (2.74)	0.446	97%

Table 10: Combined model with one reading at 3 pm, 6 pm and 9 pm (200 random splits).

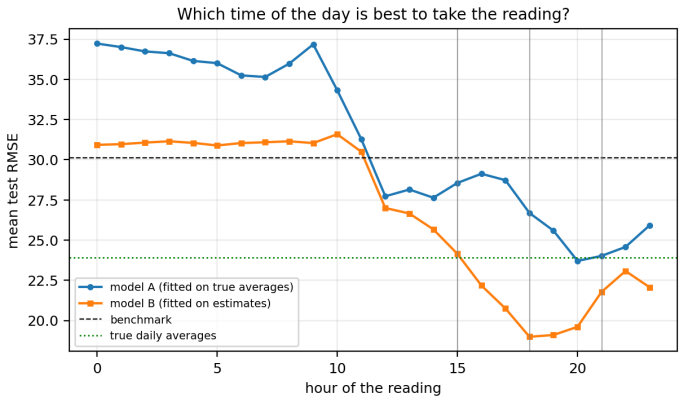


Figure 13: Mean test RMSE of models A and B for a reading taken at each hour of the day.

Which time is best? For model B the best time of the three is **6 pm**: RMSE 19.0 and $R^2_{\text{oos}} = 0.58$, followed by 9 pm (0.45) and 3 pm (0.33). Taking the 6 pm reading instead of 3 pm gives a lower RMSE in 94% of the splits (mean difference -5.2), and instead of 9 pm in 87% (mean difference -2.8). These shares are good but not overwhelming, since each test set has only 27 days, so we say 6 pm is probably the best time, not that it is sure. Figure 13 supports it: for model B the RMSE is flat at about 31 (the benchmark level) from midnight to 10 am, so a morning reading tells nothing about the day, and it goes down from noon to its minimum at 6–7 pm (19.0 and 19.1) before going up again. Model A, which is the idea of "use the daily model on estimated averages", is clearly worse than B at 3 pm and 6 pm, and it is best at 9 pm ($R^2_{\text{oos}} = 0.32$), where it almost reaches the daily model with true averages (0.34). Estimating the daily averages is easiest for the temperatures: the R^2 between the reading and the daily mean of the same variable is 0.82–0.99 for the temperatures at the three times (0.99 for `T9`), but it is poor for `lights` (0.22–0.38), `RH_5` (0.18–0.42) and `Visibility` (0.02–0.25).

Model B at 6 pm is even better than the model that uses the true daily averages of the whole day (0.58 against 0.34). We did not test why, but a plausible reason is that the conditions in the evening are partly the result of how much the appliances were used during that day, so an evening snapshot carries information that the all day average washes out. This also means that it is an association and not a forecast: the reading comes from the same day we predict. We also tried only random splits for this part, and we picked the three times before looking at results, but the hour curve was computed for all 24 hours, so the exact numbers are a bit optimistic.

4 Conclusion

- The plain regression on the 10 minute data does not work ($R^2_{\text{oos}} = 0.09$). Its errors are strongly autocorrelated ($D = 0.62$), their variance grows with the fitted value, they are far from normal, and the covariates are very collinear.
- An AR(14) model predicts the next 10 minutes well (RMSE 60 against 87) but needs 140 minutes of past data, looks only at the past of `Appliances`, and after a few hours it is no better than the mean.
- Averaging by day removes most of the dependence (r_1 from 0.69 to 0.08) and the heavy tail. Variance is still not exactly constant and collinearity is stronger, so model selection is needed. Backward selection with BIC (checked against all possible regressions) gives a 7 variable model: daily means of `T3`, `T4`, `T7`, `T9`, `RH_1`, `RH_7` and `lights`, with adjusted $R^2 = 0.478$.
- The out of sample R^2 of this model is 0.34 for random splits and 0.20 in time order, and it beats the benchmark in 93% of the random splits. The model with all 25 variables is much worse (0.21 and -0.04).

- If the house reports only one reading a day, a reading at 6 pm gives the best prediction of the daily energy ($R^2_{\text{oos}} = 0.58$), better than 9 pm and 3 pm.
- **Comparison with the original paper.** Candanedo et al. [1] report an out-of-sample R^2 of 0.16 for multiple linear regression and 0.57 for gradient boosting, the best of their methods. Our plain regression on the 10 minute data is even lower (0.09), but after averaging by day and selecting variables, the 7 variable linear model reaches 0.34 under random splits, which is more than twice the 0.16 of their linear regression. With one reading at 6 pm, model B reaches 0.58, about the same as gradient boosting (0.57), and it does so with a linear model of only 7 predictors whose coefficients can be read and checked (Table 7), whereas gradient boosting is a black box. This comparison is only indicative: we predict the daily average and use our own splits, while they predict the 10 minute readings.

5 Further work

Modelling the variance. Constant variance was the weak point of the daily model (Breusch–Pagan p-value 0.012). Instead of assuming $\text{Var}(\varepsilon_d) = \sigma^2$, the variance can be taken as a function of some covariates, for example $\text{Var}(\varepsilon_d) = \sigma^2 \exp(\gamma_0 + \gamma_1 z_{d1} + \cdots + \gamma_q z_{dq})$, where the z's can be the fitted value or some of the daily covariates. The residuals (the "noise") can be used to explain this variance: regress $\log e_d^2$ on the z's, use the fitted variances $\hat{\sigma}_d^2$ as weights in weighted least squares, and iterate until the estimates settle (feasible generalized least squares). This is a more careful version of the check we did with variance proportional to the square of the mean, and it would also give standard errors that can be trusted.

A better first step in model B. In model B we estimate each daily average from the reading at time t_0 by a straight line. Since every covariate has a daily cycle (temperatures rise and fall during the day), we can model this cycle with sine and cosine terms of period 144 readings:

$$x_{d,t,j} = m_{dj} + \sum_{k=1}^{K} \left(a_{jk} \cos \tfrac{2\pi k t}{144} + b_{jk} \sin \tfrac{2\pi k t}{144} \right) + \eta_{d,t,j},$$

where m_{dj} is the daily average of day d and a_{jk}, b_{jk} describe the typical shape of the day. The coefficients a_{jk}, b_{jk} are estimated from the training days, where all readings are known. For a new day with only the reading at t_0 we estimate $\widetilde{x}_{dj} = x_{d,j}(t_0) - \sum_k \left(\hat{a}_{jk} \cos \tfrac{2\pi k t_0}{144} + \hat{b}_{jk} \sin \tfrac{2\pi k t_0}{144} \right)$, that is, the reading corrected for the typical position of the daily cycle at that time. The number of harmonics K can be chosen by cross validation. Because the shape is shared by all days, this keeps the idea of using one reading, but it uses the whole daily pattern instead of one fitted line for each variable and each time.

Basic improvements. Some simple changes need no new theory: (i) using the log of the daily energy (Box–Cox gave an interval containing 0) together with the weighted fit above; (ii) using the readings at two or three times of the day (for example 3 pm, 6 pm and 9 pm) together in the estimate of the daily averages; (iii) adding the day of the week as a predictor, since the residual ACF of the daily model had a small bar at lag 7; and (iv) testing the model on more houses or a longer period, because we only have one house and 4.5 months (136 days), and the test sets are small (27 or 56 days). Also, all our results are associations: the daily conditions are partly a result of the appliance use, so none of this is a forecast of future days or a cause and effect statement.

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[3] Brockwell, P.J., Davis, R.A. (2016). *Introduction to Time Series and Forecasting*, 3rd edition. Springer.

[4] UCI Machine Learning Repository, *Appliances Energy Prediction* data set. `https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction`

[5] Python notebook of this project: [link to be added].